SmartSense Route Register

All 47 routes the Portal registers, grouped by the order we intend to build them and why that order. Addresses, page policy and protection are extracted from the prototype source.

EXTRACTED	SOURCE	SEQUENCED AGAINST
4 September 2026	Portal prototype, `portalRoutes()`	Document 00 §5 and document 01 §7

IN RELEASE 1

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1 The build order, and why

The prototype's own implementation status is not useful to us: we are rebuilding all of it, so "already implemented" says nothing about our work. What matters is the order we take the routes in, and the dependency or risk that puts each one there.

Six stages inside document 00 §5 and document 01 §7. Their §7 names Overview, Fleet, Gateways, Geofence Activity, Organizations, RV Owners and Audit Log as the core operational set, and leaves Sensor Data, Users, Access Profiles, RV Models and Geofence Sites unplaced. We place those, and say why.

S1 FOUNDATION	Shell, route registry, deep links and alias handling, session state, active organization and the global switcher, access-context invalidation, page policy enforcement with direct-route refusal, and the seven failure states. The windowed grid component is built here as a standalone piece, before any page depends on it, because eleven surfaces do and it is the least proven part of the Jaspr stack.
S2 FLEET SPINE	Fleet and the five VIN subpages. Taken first among the feature surfaces because it exercises everything the foundation claims to provide at once: the windowed grid at scale, six facets, server-side search, three sensitive field gates, and the Gateway Replacement Transaction as the one approved mutation. If the foundation is wrong, it is wrong here, and it is cheaper to find that out on one surface than on eleven.

S3 ACCESS ADMINISTRATION	Organizations, Users and Access Profiles with their detail and Effective Access routes. Pulled ahead of the remaining read surfaces because the access matrix built in S1 is not testable until something can create organizations, invite users and assign profiles. Three routes behind one workspace presentation, each keeping its own policy row.
S4 OPERATIONAL SURFACES	Overview, Gateways, Geofence Activity, RV Owners, Audit Log, plus Geofence Sites and RV Models administration. Mostly composition over the grid and the projections S1 to S3 established. RV Models sits here rather than later because the tire position set it defines is what the alerts work in S6 validates against.
S5 SENSOR DATA	Readings, Fields, Field Detail, Charts and the privileged Raw surface. Separated from S4 because charts need server-prepared series and bucketing rather than the row grid, and because Raw is the only route in the registry that is Read for one profile and No Access for the other four.
S6 ALERTS	The operational Alerts destination, Alert Event Detail, Significant-G Detail, and the Admin authoring pair. Last of the six because it depends on the governed measurement catalogue, on RV Model tire positions, and on the event and delivery contracts being approved. Designable throughout, testable only at the end.

Thirteen routes stay out. Six are deferred or future by Product Owner decision, and six of those are the Subscriptions group plus Subscription Plans and the Admin API surface. Seven are registry only after the controlled reset of 31 August: their policy rows are preserved, the navigation no longer offers them, and the router refuses them.

2 The register

R Read **RW** Read-Write **N** No Access **NA** Not Applicable to the web channel

◆ also reaches the Owner Application

Profile order: SmartSense Administrator · OEM Manager · OEM Worker · Dealer · RV Owner

ROUTE ID	PAGE	ADDRESS	SSA	OEM M	OEM W	DLR	OWN	PROTECTED ACTIONS · FIELD GATES
S1 · FOUNDATION								1 route
shell	Shell › Global shell	/	R	R	R	R	NA	tenant.switch

ROUTE ID	PAGE	ADDRESS	SSA	OEM M	OEM W	DLR	OWN	PROTECTED ACTIONS · FIELD GATES
S2 · FLEET SPINE								6 routes
fleet	Fleet	/fleet	R	R	R	R	NA	—
vin.current ◆	Fleet › VIN Detail › Current data	/vin/:vin	R	R	R	R	NA	loc
vin.data ◆	Fleet › VIN Detail › All data	/vin/:vin/data	R	R	R	R	NA	raw
vin.history ◆	Fleet › VIN Detail › History & events	/vin/:vin/history	R	R	R	R	NA	raw loc
vin.device	Fleet › VIN Detail › Device & config	/vin/:vin/device	RW	RW	RW	RW	NA	device.command.execute gw
vin.ownership	Fleet › VIN Detail › Ownership	/vin/:vin/ownership	R	R	R	N	NA	owner
S3 · ACCESS ADMINISTRATION								6 routes
admin.orgs	Admin › Organizations	/admin/orgs	RW	RW	N	N	NA	—
admin.users	Admin › Users	/admin/users	RW	RW	N	N	NA	users.access.manage security.admin
admin.roles	Admin › Roles & Permissions	/admin/roles	R	R	N	N	NA	profile.administer
admin.users.detail	Admin › User Detail	/admin/users/:userId	RW	RW	N	N	NA	users.access.manage security.admin
admin.roles.profile	Admin › Access Profile Detail	/admin/roles/:profileId	R	R	N	N	NA	profile.administer
admin.roles.access	Admin › Effective Access	/admin/roles/effective-access	R	R	N	N	NA	owner
S4 · OPERATIONAL SURFACES								11 routes
overview	Overview	/overview	R	R	R	R	NA	—
owners	RV Owners	/owners	R	R	N	N	NA	owner
gateways.all	Gateways › All	/gateways/all	R	R	R	R	NA	gw
gateways.offline	Gateways › Offline	/gateways/offline	R	R	R	R	NA	gw
gateways.updates	Gateways › Pending Updates	/gateways/updates	R	R	R	R	NA	gw
geo.events	Geofence Activity › Events	/geo/events	R	R	R	R	NA	loc
geo.sites	Geofence Activity › Sites	/geo/sites	R	R	R	R	NA	—
admin.rvmodels	Admin › RV Models	/admin/rvmodels	RW	RW	N	N	NA	—
admin.geosites	Admin › Geofence Sites	/admin/geosites	RW	RW	N	N	NA	—
admin.audit	Admin › Audit Log	/admin/audit	R	R	N	N	NA	sensitive.export
owners.detail	RV Owners › Ownership Detail	/owners/:ownershipId	R	R	N	N	NA	owner

ROUTE ID	PAGE	ADDRESS	SSA	OEM M	OEM W	DLR	OWN	PROTECTED ACTIONS · FIELD GATES
S5 · SENSOR DATA								5 routes
sensor.readings	Sensor Data › Readings	/sensor/readings	R	R	R	N	NA	raw loc
sensor.fields	Sensor Data › Fields	/sensor/fields	R	R	R	N	NA	—
sensor.charts	Sensor Data › Charts	/sensor/charts	R	R	R	N	NA	raw loc
sensor.raw	Sensor Data › Raw	/sensor/raw	R	N	N	N	NA	raw loc
sensor.fields.detail	Sensor Data › Field Detail	/sensor/fields/:fieldId	R	R	R	N	NA	—
S6 · ALERTS								5 routes
alerts ◆	Alerts	/alerts	R	R	R	R	NA	gw
admin.alertdefs	Admin › Alert Definitions	/admin/alertdefs	RW	RW	N	N	NA	—
alerts.event ◆	Alerts › Alert Event Detail	/alerts/:alertEventId	R	R	R	R	NA	recip raw gw
admin.alertdefs.detail	Admin › Alert Definition Detail	/admin/alertdefs/:alertDefinitionId	RW	RW	N	N	NA	—
event.sigg ◆	Alerts › Significant-G Event Detail	/event/:eventId	R	R	R	R	NA	raw loc gw
OUT OF SCOPE · DEFERRED BY PRODUCT OWNER DECISION								6 routes
subs.all	Subscriptions › All	/subs/all	R	R	N	N	NA	—
subs.included	Subscriptions › Included Period	/subs/included	R	R	N	N	NA	—
subs.ownerpaid	Subscriptions › Owner-Paid	/subs/ownerpaid	R	R	N	N	NA	—
subs.usage	Subscriptions › Usage	/subs/usage	R	R	N	N	NA	—
admin.subplans	Admin › Subscription Plans	/admin/subplans	RW	N	N	N	NA	—
admin.api	Admin › API	/admin/api	RW	RW	N	N	NA	—
OUT OF SCOPE · REGISTRY ONLY, NAVIGATION WITHDRAWN								7 routes
admin.sensorcfg	Admin › Sensor Configurations	/admin/sensorcfg	RW	R	N	N	NA	—
admin.rules	Admin › Alert Rules	/admin/rules	RW	RW	N	N	NA	—
admin.notifs	Admin › Notifications	/admin/notifs	RW	RW	N	N	NA	recip
admin.firmware	Admin › Firmware	/admin/firmware	RW	N	N	N	NA	—
admin.distribution	Admin › Distribution	/admin/distribution	RW	N	N	N	NA	distribution.deploy
admin.rules.detail	Admin › Alert Rule Detail	/admin/rules/:ruleId	RW	RW	N	N	NA	—
admin.notifs.detail	Admin › Notification Detail	/admin/notifs/:notificationId	RW	RW	N	N	NA	recip

RV Owner is Not Applicable on all 47 rows. That is a web-channel decision, not a withdrawal of rights: the Owner product experience is the Owner Application, and six routes carry evidence that reaches it.

> DISCREPANCY FOUND WHILE EXTRACTING
>
> The Permission Model document says 46 routes and 230 cells. The registry holds **47 and 235**, with per-profile totals of 30 / 17, 30 / 13 / 4, 19 / 1 / 27 and 14 / 1 / 32. The source explains it: the clean Alerts module retired three route rows and minted four, and a comment records the count moving from 46 to 47 while two others still assert 46. The policy is fine; the quoted count is stale. Raised with the customer in the handoff review.

3 Protection: seven actions, five field gates

Page policy answers whether a principal may open a surface. It does not answer whether they may perform a particular act on it, or see a particular value. Those are two further layers, and both are server-resolved. Read-Write on a page never implies a protected action, and Read never implies a sensitive field.

Protected actions

ACTION	WHAT IT GUARDS	RELEASE 1
`tenant.switch`	Moving the session between authorized organization contexts. Being authorized for several is a different fact from being allowed to move between them.	Available
`users.access.manage`	Granting or withdrawing another principal's profile and scope. Viewing, inviting and suspending a user are ordinary page management.	Available
`security.admin`	Ending another principal's authenticated sessions. Session evidence is readable with the page; ending one is not.	Available
`profile.administer`	Creating, editing and retiring custom access profiles. Reading the profile list is ordinary page access.	Available
`sensitive.export`	Taking audit evidence or precise location past the Portal boundary. Ordinary list exports are not gated here.	Available
`distribution.deploy`	Approving, starting, pausing or resuming a deployment. Reading Distribution is not authority to deploy from it.	Available

ACTION	WHAT IT GUARDS	RELEASE 1
`device.command.execute`	Acting on hardware in a customer's vehicle rather than on a record. Gateway Settings, Force Push, Set Parameter, reboot, reset and direct firmware apply.	**Declared, unavailable**

The last row matters for the rebuild: no profile holds it, no endpoint accepts it, and no control for it exists. Our implementation should keep it that way rather than build the control and hide it, because a hidden button is not authorization.

Sensitive field gates

GATE	VALUES IT WITHHOLDS
`owner.identity.view`	Owner name, contact detail and account reference on ownership records and alert-recipient evidence.
`location.precise.view`	Latitude and longitude on VIN history, Significant-G evidence, geofence crossings and sensor readings.
`telemetry.raw.view`	Pre-conversion device values, including raw accelerometer axes.
`gateway.identity.view`	Device identifier, IMEI, SIM and pairing reference on Gateway records and alert evidence.
`alerts.recipient.identity.view`	Recipient name and address on historical notification-attempt evidence.

A withheld field is labelled as withheld. It is never blanked in a way that lets the reader infer the value from a neighbouring one, and never reconstructed from another field. One consequence is worth carrying into our column model: on the Significant-G grid, the acknowledgement and notification columns become unsortable when the recipient gate is closed, because row order would disclose the value.

4 Data grids and their query contracts

Eleven surfaces are server-driven grids. Each declares its filter vocabulary, its facets, its default sort and whether it takes free text. This is the closest thing the prototype has to a backend query specification, and it is what our Serverpod query endpoints have to satisfy.

GRID	FACETS	DEFAULT SORT	SEARCH
Fleet	Model · Product Mode (Gateway, Direct TPMS) · Health (Healthy, Needs Attention, Unknown) · Reporting status (as expected, overdue, Unknown) · Active conditions (None, One or more) · Configuration (Current, Pending, Drift, Failed)	`vin`	Yes
RV Owners	Retail claim (Unclaimed, Claimable, Awaiting activation, Claimed, Released) · Owner activation (Not activated, Awaiting activation, Active, Released) · Verification (Verified, Unverified, Suspended, Anonymized, Not applicable) · Product Mode	`attention`	Yes
Gateways	Reporting state · Connectivity evidence · Convergence · Model · Battery · Organization	—	No
Sensor readings	Category · Product Mode · Value state	`time`	Yes
Sensor fields	Category · Product Mode applicability · Field authorization (Not restricted, Capability-restricted)	`cat`	Yes
Geofence sites	Location type · Geometry · Site state · Address state · Organization	`name`	Yes
Geofence events	Direction · VIN · Site · Location quality · Time · Dwell · Organization	—	No
Organizations	Type · Status	—	No
Audit log	When · Actor · Action · Target type · Actor type · Result · Organization	—	No
RV Models	Product mode · Tire layout · RV type · Status	`name`	Yes
Notification activity	State · Channel · Recipient class · Attempt type · Definition	—	No

Three things in these specs are requirements rather than implementation detail, and they should survive into our contracts. Facet counts are resolved server-side over the whole authorized population, not over the page in hand. The *Active conditions* facet on Fleet is derived, meaning no record carries the field and the facet projects the bucket the predicate tests. And Fleet deliberately has no composite attention ranking, because weighting conditions against reporting against configuration is a product decision nobody has made.

Fleet columns, and an open question attached to them

The Fleet grid ships eight columns: VIN, Model, Product Mode, Health, Reporting status, Active conditions, Configuration and Last report. Four columns named in the controlled merge, Model Year, Status Age, Mileage and Location, are absent. Whether that substitution was deliberate is open Product Owner decision 12, and the request to add Organization as a display-only ninth column is deferred to the same decision. We should not resolve it by picking a column set.

5 Behaviour to build once, not per screen

Most of the work is not 47 screens. It is a few shared mechanisms every screen then uses.

The windowed grid

The server owns the window. The client sends query intent and never hands the service pre-filtered or pre-sorted rows. Windows are 200 rows, the client retains four of them, so the memory ceiling is 800 records whether the authorized population is 400 or 100,000. Cursors are opaque, encoded as `cur:` followed by a zero-padded offset. Row height, viewport height and overscan are fixed and the window index is derived in exactly one place, because the planner and the scroll guard disagreeing about which row an offset starts at is what produced the reported scrolling fault.

Seven failure states, each with its own copy

STATUS	WHAT THE PORTAL SAYS
Unauthenticated	Not signed in. Serverpod resolved no principal for the request.
Forbidden	Not available to this principal. The result discloses nothing about what exists beyond your authorization.
NotFound	No such object in your authorized scope. A direct route is not a way around authorization, and the response does not reveal whether the record exists.
Validation	Change not accepted. Serverpod rejected the request as invalid. Nothing was applied.
Conflict	Changed by someone else. Nothing was overwritten. Refresh and review before resubmitting.
Unavailable	Serverpod could not be reached. No state was changed, and nothing shown here is a result.
Unexpected	The request failed for an unrecognised reason. No change was applied.

Forbidden and NotFound read alike from outside by design. Every response carries a correlation id and the contract version, so a Portal failure joins to its audit record on the server.

Addressability

Every route reloads, bookmarks and shares. Six child records are addressable inside their parent route rather than as routes of their own: gateway detail, geofence site detail through both the operational and administrative paths, audit event detail, RV Model detail and organization detail. Reading the page authorizes reading a row of it, so these need no separate policy row. A set of non-canonical URL aliases is accepted on read and rewritten to the canonical form, so links shared from earlier builds keep working while the emitter converges on the registry pattern. We should carry that alias table forward rather than break existing links.

Active organization context

One organization is resolved at sign-in from the principal's home organization. There is no chooser, and no feature page asks which organization. Changing context goes through the global switcher, requires `tenant.switch`, and invalidates open editors, selections, query windows, cursors, facets, counts and cached data. The server advances an access-context version so a stale response cannot be reused. In the prototype an unresolved active organization means an invalid session, never a prompt to pick one.

Internal working document. Extracted from the Portal prototype's own route registry, access policy, query specifications and error vocabulary on 4 September 2026, so the register reflects the source rather than the documents that describe it. Where the two disagree, section 1 records the difference.